Laporan Pengujian Program Level Tangki

1. Tabel Pengujian (Test Case)

Pengujian dilakukan untuk memastikan fungsi program berjalan sesuai ketentuan kategori level tangki.

Test Case	Deskripsi	Expected Output	Actual Output	Status
TC01	Level 18%	LOW	LOW	**PASS**
TC02	Level 29%	LOW	LOW	**PASS**
TC03	Level 30%	NORMAL	NORMAL	**PASS**
TC04	Level 80%	NORMAL	NORMAL	**PASS**
TC05	Level 81%	HIGH	HIGH	**PASS**
TC06	Level 91%	HIGH	HIGH	**PASS**
TC07	Menghitung jumlah kategori	LOW: 5, NORMAL: 11, HIGH: 4	LOW: 5, NORMAL: 11, HIGH: 4	**PASS**
TC08	Menghitung rata-rata 20 data	50,85%	50,85%	**PASS**

2. Kode Testing Program

Tambahkan kode berikut di bagian paling bawah program Python Versi 1, setelah seluruh fungsi dan data dibuat.

```
print("\n====================================")
print("       TESTING PROGRAM")
print("====================================")

test_cases = [
    ("TC01 - Level 18%", "LOW", status_level(18)),
    ("TC02 - Level 29%", "LOW", status_level(29)),
    ("TC03 - Level 30%", "NORMAL", status_level(30)),
    ("TC04 - Level 80%", "NORMAL", status_level(80)),
    ("TC05 - Level 81%", "HIGH", status_level(81)),
    ("TC06 - Level 91%", "HIGH", status_level(91)),
    ("TC07 - Jumlah kategori",
     (5, 11, 4), hitung_kategori(data_level)),
    ("TC08 - Rata-rata",
     50.85, round(rata_rata(data_level), 2))
]

lulus = 0

for nama, expected, actual in test_cases:
    status = "PASS" if expected == actual else "FAIL"

    if status == "PASS":
        lulus += 1

    print(f"{nama}")
    print(f"Expected : {expected}")
    print(f"Actual   : {actual}")
    print(f"Status   : {status}")
    print("------------------------------------")

print(f"Total Testing : {len(test_cases)}")
print(f"PASS          : {lulus}")
print(f"FAIL          : {len(test_cases) - lulus}")
print("====================================")
```

3. Bukti Output Terminal

Berikut bentuk output terminal yang akan dihasilkan jika seluruh pengujian berhasil:

```
====================================
       TESTING PROGRAM
====================================

TC01 - Level 18%
Expected : LOW
Actual   : LOW
Status   : PASS
------------------------------------

TC02 - Level 29%
Expected : LOW
Actual   : LOW
Status   : PASS
------------------------------------

TC03 - Level 30%
Expected : NORMAL
Actual   : NORMAL
Status   : PASS
------------------------------------

TC04 - Level 80%
Expected : NORMAL
Actual   : NORMAL
Status   : PASS
------------------------------------

TC05 - Level 81%
Expected : HIGH
Actual   : HIGH
Status   : PASS
------------------------------------

TC06 - Level 91%
Expected : HIGH
Actual   : HIGH
Status   : PASS
------------------------------------

TC07 - Jumlah kategori
Expected : (5, 11, 4)
Actual   : (5, 11, 4)
Status   : PASS
------------------------------------

TC08 - Rata-rata
Expected : 50.85
Actual   : 50.85
Status   : PASS
------------------------------------
Total Testing : 8
PASS          : 8
FAIL          : 0
====================================
```